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Study on CaO-promoted Co/AC catalysts for synthesis of higher alcohols from syngas

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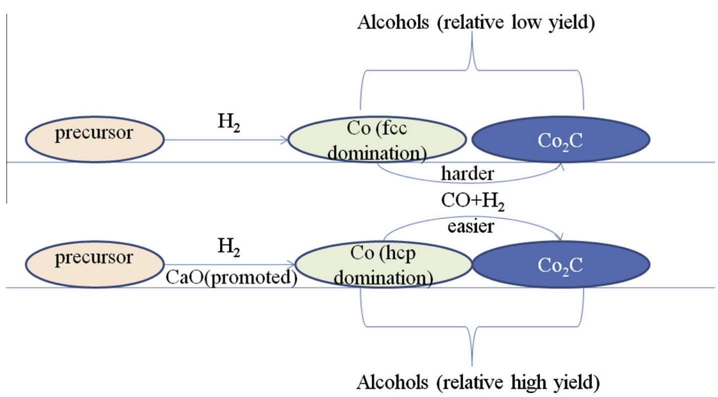
HIGHLIGHTS

- Co–CaO/AC catalysts were prepared and used for the production of mixed higher alcohols.
- The CaO facilitated the generation of the hcp Co species during the activated process.
- The CaO species promoted the development of the Co_2C species during the reaction.
- The 15Co0.1Ca/AC catalyst shows the excellent performance for the alcohols synthesis.
- The synergistic effect of Co^0 and Co_2C are responsible for the alcohols synthesis.

GRAPHICAL ABSTRACT



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ABSTRACT

An alternative CaO-promoted Co/AC catalysts with high selectivity of C_1–C_{16} alcohols were synthesized and systematically studied. The catalysts were characterized by means of N_2-physisorption, XRD, H_2-TPR, CO-chemisorption and HRTEM. Then the activated catalysts were used in the CO hydrogenation reaction for the alcohols synthesis. Reaction results showed that CaO promoter improved the yield of C_1–C_{16} alcohols significantly from 9.9% to 15.0% in comparison with un-promoted Co/AC catalyst. Characterization results indicated that the addition of CaO promoted the formation of hcp metallic Co, which was further liable to conversion into Co_2C species. It was suggested that Co_2C species was related to synthesis of alcohols. It was found that the appropriate relative ratio of Co_2C/Co was beneficial to synthesis of mixed C_1–C_{16} alcohols over the CaO-promoted Co/AC catalysts. The synergistic effect of metallic Co nanoparticles and Co_2C species, which contact intimately, is responsible for the alcohols synthesis.

1. Introduction

Ethanol and higher alcohols have been considered as potential fuels/fuel additives ($C_nH_{2n+1}OH$, $n = 2–5$) [1] and feedstock for high value chemicals including plasticizers, detergents and lubricants, etc. ($C_nH_{2n+1}OH$, $n \geqslant 6$) [2]. With the depletion of traditional petroleum reserves and the increasing concern of global environmental

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issue, a promising route to produce higher alcohols from syngas ($CO + H_2$) derived from coal, natural gas or biomass is Fischer–Tropsch (FT) synthesis [3].

In the past decades, four types of catalysts have been developed for the synthesis of higher alcohols from syngas, including Rh-based catalysts, modified FT catalysts, Mo-based catalysts and promoted methanol catalysts [4]. Rh-based catalysts are highly selective to ethanol, and the alkali/alkaline earth oxides were proposed as the appropriate promoter [5]. CuCo-based catalysts were mainly developed as the modified FT catalysts for synthesis of higher alcohols. The selectivity for alcohols on the catalysts depends on the type of promoters used. The interaction between highly dispersed Co and Cu metallic particles is important in catalytic activity for higher alcohols synthesis [6–8]. MoS_2 catalysts were mostly used among the Mo-based catalyst for higher alcohols synthesis. The alcohols selectivity and yield increased with the promotion of alkali metals. Besides, the group VIII elements were the suitable promoter for the Mo-based catalysts [9]. In addition, the alkali/alkaline earth metals were added into the methanol catalysts, such as CuZnAl and $ZnCrO_x$, in order to modify the performance of alcohols synthesis [10,11]. Recently, alkali metals modified cobalt catalysts were reported for the alcohols synthesis [12]. Although great efforts have been made in the field of higher alcohols synthesis based on the four types of catalysts aforementioned in order to enhance the alcohols selectivity and yield, the selectivity of long-chain alcohols (especially C_{6+} alcohols) over these catalysts is still not high enough [1].

We have explored an one-step method for direct synthesis of C_1–C_{18} alcohols together with high quality fuels and low methanol via FT reaction using abundant cobalt catalysts supported on an activated carbon carrier [13]. The major characteristic of these catalysts is that during catalytic reaction, Co metallic nanoparticles can be partly carbonized by CO into Co_2C, forming a dual active center including Co metal and Co_2C phase which can increase carbon chain length and insert CO molecules into carbon chains, respectively [14].

It is known that addition of some metal oxide promoters to cobalt catalysts can modify the support's texture, regulate the Co particle size and modulate Co reducibility, so that the FT activity and selectivity can be improved [15]. The alkali and alkaline earth metals have been used as promoter for Co or Fe-based catalysts in the FT synthesis, especially for alcohols synthesis catalysts for the purpose of improving the selectivity towards alcohols and alcohols yield [16–19]. The improvement effect of alkali and alkaline earth metals on the Co catalysts resulted from the greater abundance of Co^{2+} species and the increase in surface basicity. We have previously shown that the moderate amount of K and Li doping was beneficial to the alcohols synthesis over the Co/AC catalysts [20,21]. Besides, alkali metals seem to be able to promote the formation of Co_2C species [22]. Co_2C species was known as the necessary active sites for the alcohols synthesis for monometallic cobalt catalysts [13,14,23,24].

In this work, the alkaline earth metal oxide CaO promoted Co/AC catalysts for synthesis of C_1–C_{16} alcohols from syngas will be investigated in order to achieve high selectivity and yield of C_1–C_{16} alcohols. To the best of our knowledge, no study related to the direct synthesis C_1–C_{16} alcohols together with hydrocarbons over Co–CaO/AC catalyst system has been reported.

2. Experimental

2.1. Preparation of catalysts

An activated carbon (labeled as AC, Brilliant Tech Co. Ltd.) was made from coconut shells. The pristine AC was washed with deionized water until ca. pH = 7, then crushed into 40–60 mesh. The AC supported un-promoted Co catalyst was prepared by the incipient wetness impregnation method according to the following procedure: the AC was impregnated at room temperature with an aqueous solution of $Co(NO_3)_2 \cdot 6H_2O$. Then the samples were dried in air at 333 K for 12 h. The CaO-promoted catalysts were prepared by impregnating the AC with mixed solution of cobalt nitrate and calcium nitrate, and dried under the same conditions as described above. The dosage of the CaO was selected based on our previous studies, in which the trace amount of alkali metals was proved to be appropriate to modify the Co/AC catalyst with higher catalytic performances [20,21]. The un-promoted and CaO-promoted catalysts were denoted as 15Co/AC and 15Co–xCa/AC (x = 0.05, 0.10 and 0.50), respectively.

2.2. Evaluation of the CO hydrogenation reaction

CO hydrogenation reaction was performed in a stainless fixed-bed micro-reactor with an inner diameter of 9 mm. In each test, 2 ml of catalysts diluted with 2 ml of quartz were loaded into the reactor. Prior to the reaction, the catalysts were in situ reduced at 703 K for 10 h in a flow of H_2 (900 h^{-1}) at 0.1 MPa. After the samples were cooled to 353 K, the syngas ($H_2/CO = 2$, GHSV = 900 h^{-1}, P = 3.0 MPa) was fed into the reactor. Then, the furnace was heated to the reaction temperature in two stages, firstly from 353 to 473 K, then to 493 K with ramping rates of 2 K/min and 0.1 K/min, respectively. The reaction effluent successively passed through two traps, one maintained at 373 K for the collection of the product with high boiling point and the other at 273 K for the gather of the volatile component. The data were all taken after 36 h of stabilization. The detailed analysis methods of the products could be found in the literature [13].

2.3. Characterization of catalysts

The specific surface area, pore volume and average pore size of AC carrier and calcined 15Co–xCa/AC catalysts were evaluated using N_2-physisorption at 77 K on an Autosorb-1 instrument. 30 mg of each sample was out-gassed under vacuum at 473 K for 3 h prior to the measurement.

Power X-Ray diffraction (XRD) phase analysis was carried out with a PANalytical X'Pert^3 Powder diffractometer using CuKα radiation. The test was operated at 40 kV and 40 mA. The fresh catalysts and spent catalysts were measured. The identification of the different phases was made using the JCPDS library. The average particle size was calculated from XRD patterns using the Debye–Scherrer's equation.

H_2-TPR experiments were carried out on a Zeton Altramira AMI-300 unit. 100 mg of calcined sample was first flushed with argon at 393 K for 1 h and then cooled down to 323 K. Subsequently, the sample was heated to 1173 K (10 K/min) in a flow of 10% H_2/Ar mixture at 30 ml/min. TCD signal was recorded from 323 K to 1173 K. H_2-TPR tests were also conducted for the reduced catalysts (703 K, 10 h). The reduction degree was calculated based on the TPR profile of the reduced catalysts.

Pulsed CO chemisorptions experiments were carried out on the same instrument as H_2-TPR study. 100 mg of each calcined sample was in situ reduced by 10% H_2/Ar at 703 K for 4 h, purged with helium at the same temperature for 10 min to remove adsorbed species from its surface and then cooled down to 323 K. CO adsorption was carried out at 323 K by means of pulse chemisorptions with 10% CO/He until the TCD signal reached a constant value.

HRTEM measurements were carried out on a FEI Tecnai G2 F30 microscope at an accelerating voltage of 300 kV with a point resolution of 0.2 nm and line resolution of 0.10 nm.

3. Results and discussion

3.1. Catalysts characterizations

The textural parameters of the AC carrier and the AC supported catalysts with or without CaO doping are provided in Table 1 and Fig. S1. The pore passage of AC support comprises micropore and mesopore as displayed in Fig. S1. BET surface area and total pore volume dropped obviously after Co and CaO was impregnated onto the support. The average pore size of the catalysts decreased compared to the original AC support. Meanwhile, the surface area and volume of the micropore decreased slightly. These results indicated that the decrease of surface area and pore volume mainly came from the contribution of mesopore. Moreover, the average particle size of cobalt is higher than 10 nm as defined by XRD profiles. Therefore, it can be suggested that the metal primarily located at the mesopore and the external surface of activated carbon, which is identical with the report by Ma et al. [25]. This illustrated that the catalytic reaction mainly occurred in the mesopore and the external surface of AC support.

Fig. 1 displays the H_2-TPR profiles of the 15Co–*x*Ca/AC catalysts with different amount of CaO. Three peaks are observed in the TPR profiles. The broad hydrogen consumption peak at about 770 K might be ascribed to hydrogenation of the surface organic groups of the AC support [26]. The other two peaks were assigned to the two-step reduction of cobalt oxide ($Co_3O_4 \rightarrow CoO \rightarrow Co$) [27]. The samples containing CaO have slightly higher reduction temperature (626–630 K), and the more of CaO, the higher for the reduction temperature. The stronger interaction between CaO and cobalt resulted in the observation. It suggests that the existence of CaO inhibited the reduction of cobalt oxide species, especially for the drastic reduction of CoO_X species [13].

XRD patterns of the fresh catalysts are given in Fig. 2a. There is a dispersive diffraction peak (2θ = 10–30°) assigned to activated carbon in all of the reduced 15Co–*x*Ca/AC samples [28]. Two broad peaks at about 36.8° and 65.2°, which are ascribed to Co_3O_4 phase produced during the passivation process, were detected on the fresh catalysts. The diffraction peaks at 2θ = 41.5°, 44.5°, 47.4°, and 75.8° are characteristic of metallic cobalt with hcp stacking according to PDF 01-089-7094. Whereas, the characteristic peak (2θ = 44.2°, PDF 01-089-7093) of metallic cobalt with fcc phase is overlapped with the diffraction peak from hcp Co at 2θ = 44.5°. Accordingly, the fcc Co and hcp Co coexist in the reduced catalysts. Similar diffraction peaks were also found on the 15Co–*x*Ca/AC catalysts. Moreover, the intensity of peaks from hcp metallic cobalt strengthened with the introduction of CaO. This reveals that doping of CaO promoted the development of hcp metallic cobalt during reduction.

The XRD patterns of the used catalysts are shown in Fig. 2b. It is obvious that those patterns are apparently different from the patterns of the fresh samples. Besides the diffraction peaks attributed to metallic Co (fcc and hcp), some peaks at 37.0°, 41.3°, 42.6°, 45.8° and 56.6° were observed, which can be ascribed to cobalt carbide (Co_2C) according to PDF 01-072-1369. Occurrence of those diffraction peaks after reaction indicated that Co_2C species formed during CO hydrogenation under reaction conditions. HRTEM has been performed in order to verify the crystalline structure of metal in used catalysts. The clean lattice fringes of 2.25 Å and 2.21 Å as well as

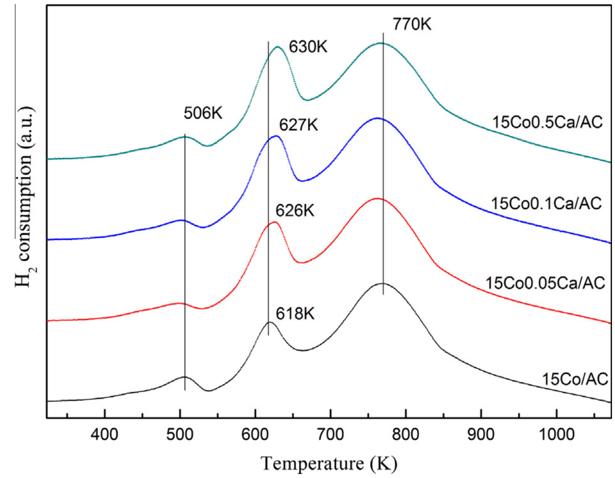

Fig. 1. H_2-TPR profiles of the 15Co–*x*Ca/AC catalysts.

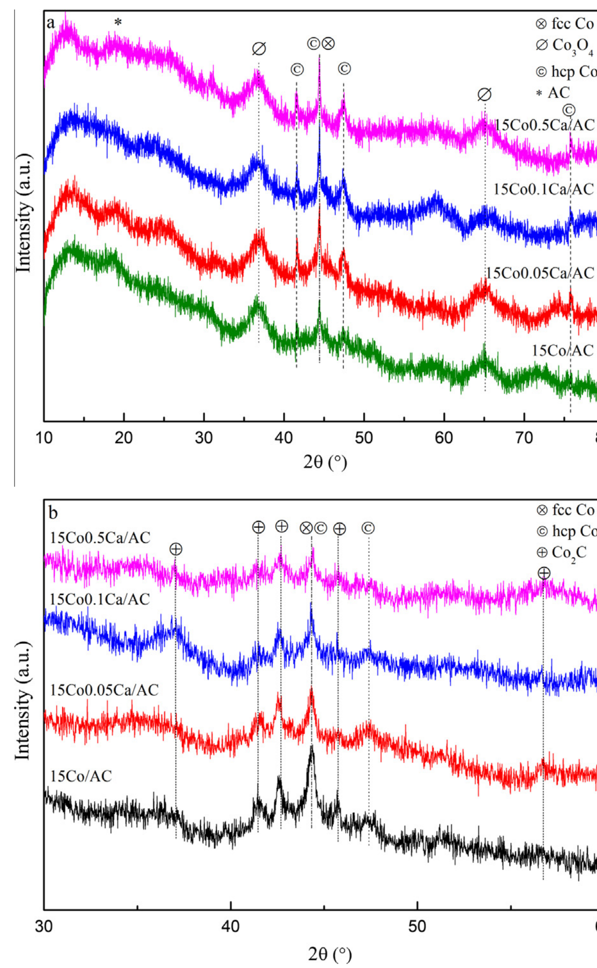

Fig. 2. XRD patterns of the fresh (a) and spent (b) 15Co–*x*Ca/AC samples.

Table 1
Textural parameters of activated carbon and corresponding catalysts.

Sample	A_{BET}/(m^2/g)	A_{micro}/(m^2/g)	V_{total}/(cm^3/g)	V_{micro}/(cm^3/g)	D_{pore}/(nm)
AC	976	468	0.70	0.20	1.4
15Co/AC	890	398	0.66	0.20	0.9
15Co0.05Ca/AC	815	429	0.56	0.19	1.1
15Co0.10Ca/AC	724	431	0.46	0.19	1.1
15Co0.50Ca/AC	695	396	0.46	0.17	1.0

A_{BET}: BET surface area. A_{micro}: micropore surface area. V_{total}: total pore volume. V_{micro}: micropore volume. D_{pore}: average pore size.

the angle of 91° between two lattice fringes are the characteristic interlayer spacing of the Co_2C (200) and Co_2C (020) planes and their corresponding angle in Fig. 3a, and fcc Co (111) plane was observed in Fig. 3a. Besides, characteristic planes of Co_2C as well as metallic fcc Co were detected in the other three spent catalysts (Fig. 3b–d). These observations proved that the metallic cobalt particles and cobalt carbide species exist simultaneously in the spent samples. The phenomenon is consistent with XRD finding and our previous studies [13,14]. From the XRD patterns of fresh and spent catalysts, it can be seen that the intensity of diffraction peaks ascribed to metallic cobalt increased with the introduction of CaO for fresh catalysts, while it decreased with the amount of CaO increased for used catalysts. Furthermore, the peaks attributed to hcp Co for spent catalysts almost disappeared in comparison with the patterns of fresh catalysts, while the peak for fcc Co remained. These observations implied that almost all of the hcp metallic cobalt converted into Co_2C species after reactions for the catalyst system. Claeys et al. [29] observed the similar phenomenon during the carburization of metallic Co. The DFT (density functional theory) calculation showed that CO activation on the hcp Co has much higher intrinsic activity than that of fcc Co [30]. And some experimental testified that the cobalt with domination hcp Co shows better CO hydrogenation performance than cobalt with mainly fcc phase [31]. Furthermore, Co_2C structure is similar to the structure of hcp metallic cobalt [32]. On the basis of the above observations, it can be speculated that hcp Co nanoparticle was liable to conversion into Co_2C species in comparison with fcc Co when cobalt catalyst was exposed to syngas atmosphere. Therefore, it could be concluded that the CaO doping promoted the formation of hcp metallic cobalt, which was further converted into Co_2C phase during the CO hydrogenation reaction.

The Co_2C/Co relative ratio, which was determined by the ratio of XRD intensity for main Co_2C peak to that for main Co peak [24], and average crystallite diameter of cobalt species for the catalysts are listed in Table 2. The Co_2C/Co relative ratio increased from 0.44 to 0.78 with the increasing of Ca loading from zero to 0.50 wt%. It indicated that the addition of CaO promoted the metallic Co conversion into Co_2C species, leading to a decrease of the quantity of metallic Co sites and an increase of the Co_2C species. With respect to the product selectivity in Table 3, it was found that a moderate relative ratio of Co_2C/Co was favorable for synthesis of mixed C_1–C_{16} mixed alcohols, which is in accord with research of Wang et al. [24]. Reduction degree, CO uptakes and corresponding cobalt dispersion of the 15Co–*x*Ca/AC catalysts are also summarized in Table 2. The reduction degree of the 15Co–*x*Ca/AC catalysts decreased slightly from 74% to 71% when Ca loading increased from 0 wt% to 0.05 wt%, but it remained almost unchanged with the further increase of Ca loading. The quantity of adsorbed CO decreased from 112.9 μmol/g to 98.5 μmol/g when the loading of Ca increased from 0 wt% to 0.50 wt%. Corresponding cobalt dispersion calculated from CO uptakes decreased with increasing of Ca content. This result is in line with the dispersion calculated from cobalt particle size derived from XRD patterns. The quantity of metallic Co sites as well as cobalt dispersion dropped with the addition of CaO, leading to the decrease of the quantity of the exposed availably metallic Co sites.

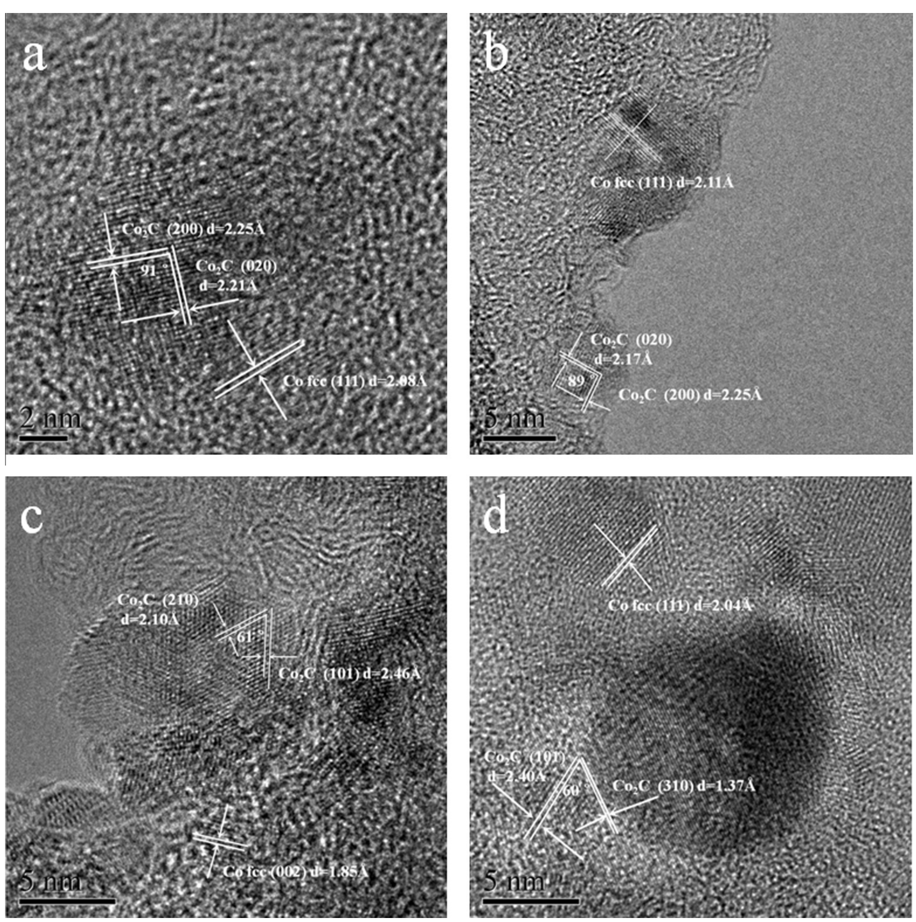

Fig. 3. HRTEM images of the spent 15Co–*x*Ca/AC catalysts: (a) 15Co/AC sample; (b) 15Co0.05Ca/AC sample; (c) 15Co0.1Ca/AC sample; (d) 15Co0.5Ca/AC sample.

Table 2
CO uptakes, Co_2C/Co relative ratio, average crystallite diameter of cobalt species, dispersion and reduction degree of metallic cobalt for 15Co–*x*Ca/AC catalysts.

W(Ca)/%	CO uptake[a] (μmol/g cat.)	Co_2C/Co[b]	D(XRD)/nm[c]			Co dispersion/%		Reduction degree/%[h]
			fcc Co[d]	Co_2C[e]	hcp Co[d]	XRD[f]	CO chemisorptions[g]	
0	112.9	0.44	18.5	25.6	9.0	5.2	4.44	74
0.05	105.4	0.50	21.2	21.1	15.3	4.5	4.14	71
0.10	103.5	0.64	26.5	21.6	13.2	3.6	4.07	71
0.50	98.5	0.78	28.3	21.1	15.6	3.4	3.87	72

[a] Based on CO pulse chemsorption.
[b] The Co_2C/Co ratio determined from the intensity of the XRD peaks characteristic of Co_2C and Co.
[c] The cobalt species particle size calculated from the XRD pattern using Scherrer equation.
[d] Fresh samples.
[e] Used samples.
[f] The dispersion of cobalt based on d (fcc Co) = 96/D%.
[g] The dispersion of cobalt based on CO/Co = 1:1.
[h] The reduction degree calculated based on the TPR profile of the reduced samples.

Table 3
The performance of CO hydrogenation over the 15Co–*x*Ca/AC catalysts with different CaO dopings.[a]

W(Ca)/%	CO conversion/%	Selectivity/C%				Alcohol yield/%	Alcohol distribution/wt%			Liquid phase products distribution/wt%	
		HC_{1-4}[b]	CO_2	HC_{5+}[c]	ROH[d]		MeOH[e]	$C_{2-5}OH$[f]	$C_{6-16}OH$[g]	Hydrocarbon	Alcohol
0	64.3	37.7	1.5	45.3	15.4	9.9	5.2	33.8	61.0	68.2	31.8
0.05	56.2	32.7	1.4	42.5	23.5	13.2	6.7	40.0	53.2	57.0	43.0
0.10	49.0	29.9	0.7	38.8	30.6	15.0	3.6	43.8	52.7	47.0	53.0
0.50	24.7	48.0	1.3	21.4	29.3	7.2	4.2	63.2	32.6	29.6	70.4

[a] Reaction conditions: $P = 3.0$ MPa, $T = 493$ K, GHSV = 900 h^{-1}.
[b] Hydrocarbons with carbon numbers of 1–4.
[c] Hydrocarbons with carbon numbers more than 4.
[d] Alcohols with carbon numbers of 1–16.
[e] CH_3OH.
[f] Alcohols with carbon numbers of 2–5.
[g] Alcohols with carbon numbers of 6–16.

3.2. Catalytic performance of CO hydrogenation

The results of reactions over the 15Co–*x*Ca/AC catalysts conducted under moderate conditions are shown in Table 3. It is evident that the hydrocarbons and alcohols were produced over all of the catalysts. Besides, minor amount of CO_2 generated. CO conversion decreased slightly from 64.3% to 56.2% when Ca doping increased from 0 wt% to 0.05 wt%, and it consistently decreased with the continued increasing of Ca content, which is in agreement with the previous reports, which suggested that the addition of alkaline earth promoter resulted in the decrease of activity for cobalt based FT synthesis catalyst [33,34]. As we know, the catalytic activity of the Co-based catalyst for CO hydrogenation depends upon the quantity of the exposed available metallic Co sites [15]. According to the characterization results, the Co dispersion calculated from the CO adsorption decreased with the increasing of Ca content, and the promotion for the formation of Co_2C with the addition of CaO led to a decrease of the Co^0 sites. Thus, the drop of CO conversion with the Ca doping is associated with the decrease in the quantity of the Co^0 active sites.

The products selectivity was also notably influenced by the addition of CaO. When Ca content increased from 0 to 0.1 wt%, the selectivity towards alcohols increased considerably from 15.4% to 30.6%, while the selectivity to HC_{1-4} hydrocarbons decreased from 37.7% to 29.9% and the selectivity to HC_{5+} hydrocarbons dropped from 45.3% to 38.8%, respectively. The results indicated that the 0.1 wt% Ca doping of the 15Co–*x*Ca/AC catalyst improved the selectivity towards alcohols by suppressing the selectivity to hydrocarbon in our study. As Ca content continued to increase to 0.5 wt%, the selectivity to alcohols remained almost unchanged, while the selectivity to HC_{1-4} hydrocarbons increased and the selectivity to HC_{5+} hydrocarbons dropped.

It can be seen that the yield of alcohols increased with the CaO doping and reached up to a maximum of 15.0% over the catalyst with 0.10 wt% Ca content. The doping of CaO into the 15Co/AC catalyst also impacted on the alcohols distribution. It is observed that the proportion of C_{2-5} alcohols in alcohols distribution monotonically increased, whereas the proportion of C_{6-16} alcohols in alcohols distribution decreased with the increase of the amount of Ca content. The distribution of liquid phase products of the 15Co–*x*Ca/AC catalysts are also listed in Table 3. It is found that the alcohols content among the liquid phase products increased with the increasing of Ca content. In particular, the content of alcohols among the liquid phase products of the 15Co0.1Ca/AC catalyst is almost 53 wt%. Based on above observations, it is suggested that the moderate CaO doping into Co/AC catalyst favored the formation of mixed higher alcohols, especially for the C_{2-5} mixed alcohols.

The product distribution curves were plotted based on ASF kinetics. The results show that the hydrocarbons and alcohols produced over the 15Co–*x*Ca/AC catalysts all obeyed ASF distribution. The representative product distributions of olefins, paraffins and alcohols are displayed in Fig. 4 over the 15Co0.1Ca/AC catalyst. It can be seen that alcohols, olefins and paraffins have almost the same chain growth probability ($\alpha = 0.66 \pm 0.01$). The ASF distributions of other catalysts are shown in Fig. 5, it is noted that α value of alcohols and hydrocarbons decreased when Ca content increased successively. This suggested that Ca doping exerted an inhibitory influence on carbon chain growth to some extent. However, α value of alcohols and hydrocarbons are still the same for all catalysts no matter the CaO doped or not. Besides, it has been confirmed that the synthesized alcohols via FT synthesis over the Co/AC catalysts are linear carbon chain [13,24]. Accordingly, it can be speculated that the chain growth of alcohols and hydrocarbons

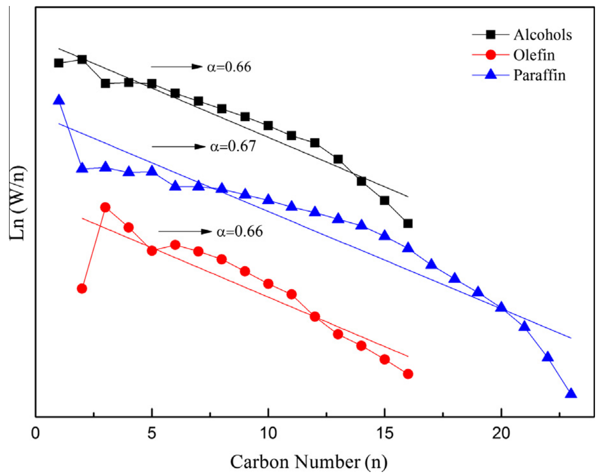

Fig. 4. ASF distributions of products over the 15Co0.1Ca/AC catalyst.

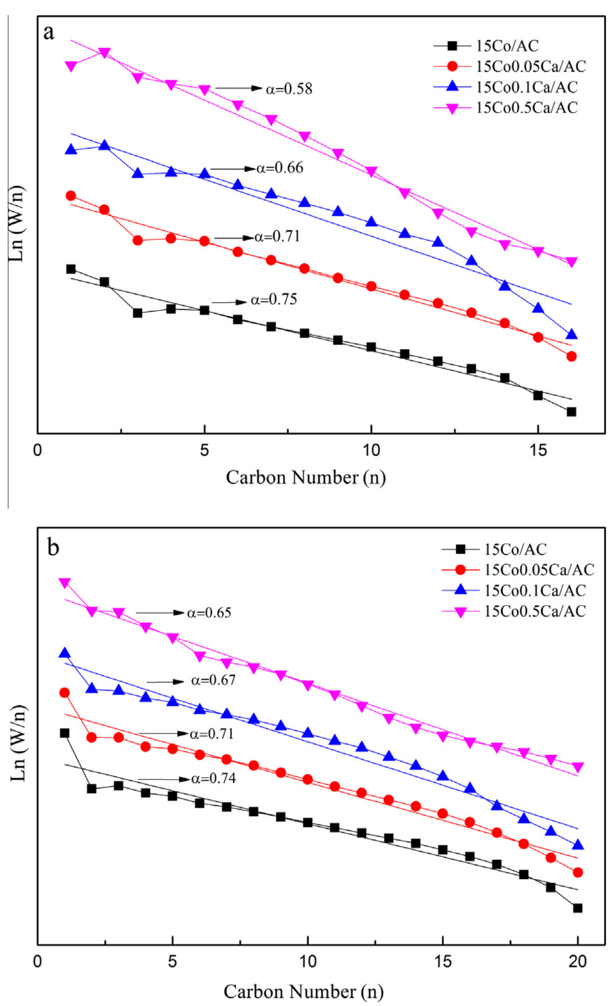

Fig. 5. ASF distributions of alcohols (a) and hydrocarbons (b) on the 15Co–*x*Ca/AC catalysts.

completed in the same active sites. Thus, the carbon chain growth was conducted on the metallic cobalt sites both for alcohols as well as hydrocarbons.

3.3. The reaction mechanism analysis

The Co_2C formation during the reaction over the Co–Cu catalysts and its vital function on the alcohols synthesis was reported by Volkova et al. [35], Fang et al. [36], Liu et al. [37]. Besides, activated carbon supported cobalt catalysts promoted by La_2O_3 [13,24], Li_2O [38] and Al_2O_3 [39] covered that the Co_2C developed via the reaction and the alcohols selectivity had close relation with the relative ratio of Co_2C and Co. Recently, our research proved that the formation of the stable Co_2C and the Co–Co_2C interface are the key for the alcohols formation [14]. We also demonstrated that the Co_2C serve as the activated sites for CO non-dissociative adsorption, whereas the Co is highly efficient for CO dissociative adsorption and in turn the carbon chain growth. The double function centers work like the bi-functional catalysts such as Fe–Cu, Co–Cu and so on [7,40]. As a consequence, it can be deduced that the formation of Co_2C species and the proper relative ratio of Co_2C/Co are the essential condition for the alcohols production over our 15Co–*x*Ca/AC catalysts.

As depicted in Section 3.2, the mixed C_1–C_{16} alcohols generated over 15Co–*x*Ca/AC catalysts. The alcohols selectivity based on carbon number reached up to around 30% when the CO conversion kept in about 50%. The CaO-promoted 15Co–*x*Ca/AC catalysts exhibited higher selectivity and yield of the mixed C_1–C_{16} alcohols compared to un-promoted Co/AC catalyst. In addition, the addition of CaO significantly facilitated synthesis of alcohols by suppressing the production of the HC_{5+} hydrocarbons, and had less influence on the production of HC_{1-4} light hydrocarbons. To see the role of AC support for the alcohols synthesis, the 15Co0.5Ca/γ-Al_2O_3 catalyst was prepared and used for CO hydrogenation reaction. The reactivity of 15Co0.5Ca/γ-Al_2O_3 and 15Co0.5Ca/AC catalysts were compared (Table S1). The alcohols selectivity of 15Co0.5Ca/AC catalyst is three times higher than of the 15Co0.5Ca/γ-Al_2O_3 catalyst. The spent 15Co0.5Ca/γ-Al_2O_3 catalyst was characterized by XRD. The characteristic diffraction peaks of metallic Co and γ-Al_2O_3 were observed in the XRD pattern (Fig. S2). The absence of the Co_2C diffraction lines for 15Co0.5Ca/γ-Al_2O_3 catalyst and the existence of the Co_2C species for 15Co0.5Ca/AC sample suggested that the γ-Al_2O_3 suppressed the formation of Co_2C species during reaction while the AC facilitated the Co_2C phase formation. This observation is in accord with the report of Wang et al. [24], which certified that AC support promoted the Co_2C formation, whereas oxides support inhibited it. Hence, the production of alcohols in our study partly resulted from the use of activated carbon (made from coconut shell) support. In addition, the fact that the Co_2C formation was also facilitated due to the presence of CaO promoter has been verified in Section 3.1. And, CaO promoted the formation of Co_2C species through facilitating production of hcp metallic cobalt during reduction.

As illustrated before, the alcohols received from our reaction are basically linear alpha ones. This fact demonstrates that the generation of alcohols via hydroformylation of primary olefin, which produce linear as well as branch alcohols [41], is eliminated. Furthermore, the alkalinity of catalysts used in our reaction is extremely weak due to the limited CaO content compared to the catalyst used in the Guerbet coupling reaction, and the AC support is approximate neutral on account of the washing process as well as high temperature calcinations (Al wt% < 0.02%). Therefore, alcohols containing branch one derived from the Guerbet coupling reaction for primary alcohols is also excluded [42]. The hydrocarbon formed by means of alcohol produced by our Co/Co_2C catalysts dehydration using activated carbon as catalyst can be ruled out since the reaction should be occurred by means of highly acidic catalysts under higher reaction temperature [43].

Based on above analysis, the C_1–C_{16} alcohols are the primary products from our reaction. In addition, the reaction path for the formation of hydrocarbon over the Co–CaO/AC catalysts is the same as the conventional cobalt catalyst. The synergistic effect between Co and Co_2C is suggested in our Co–CaO/AC catalyst system like the un-promoted cobalt catalyst [14]. Then, we attempt to give the reaction mechanism for alcohols together with hydrocarbons synthesis over the Co–CaO/AC catalysts. Co_2C species in Co–CaO/AC catalysts system serve as the CO associative adsorption sites. The metallic cobalt (both fcc and hcp phases) species are the active sites for dissociative CO adsorption and hydrogenation. Then the reaction pathway for hydrocarbon together with alcohols formation from syngas over Co–CaO/AC catalysts outlined in Fig. S3. CO dissociated readily to form surface $^*CH_x$ species on metallic cobalt sites and this surface $^*CH_x$ species formed CH_4 by direct hydrogenation. CH_3OH is generated by direct hydrogenation of molecule adsorbed CO^* on Co_2C sites. The CO^* molecule moves to an adsorbed $^*CH_x$ group and inserts between Co site and the alky group via surface migration over a short distance between metallic Co and Co_2C sites, which is further hydrogenated to form ethanol. The carbon chain growth is propagated via the coupling of $^*CH_x$ species, which leads to the forming of $^*C_nH_z$ groups. Then the $^*C_nH_z$ species hydrogenated directly to formation of hydrocarbons products. Besides, the CO^* molecule inserted into $^*C_nH_z$ species and further hydrogenated to produce the C_{n+1} linear alcohols.

4. Conclusions

The addition of CaO in the Co/AC catalyst remarkably improved the selectivity and yield of C_1–C_{16} mixed alcohols, along with a drop of CO conversion. The highest yield of C_1–C_{16} alcohols was observed over 0.10 wt% Ca promoted Co/AC catalyst. The doping of CaO promoted the formation of hcp metallic Co in the catalyst reduction process. It was preferred that hcp metallic Co was transformed into Co_2C phase rather than fcc Co in FTS reaction, thus resulting in an increase of Co_2C/Co relative ratio of the catalysts. Appropriate amount of Ca of the CaO-promoted Co/AC catalysts leads to a moderate Co_2C/Co relative ratio, which was favorable for synthesis of mixed C_1–C_{16} mixed alcohols. The existence of CaO decreased the quantity of metallic Co sites and decreased cobalt dispersion of the catalysts. The synergistic effect of metallic cobalt sites and Co_2C sites are responsible for the synthesis of C_1–C_{16} alcohols from syngas over the CaO-promoted Co/AC catalysts.

Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.fuel.2016.05.089.

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Supplementary Material for

Study on CaO-promoted Co/AC catalysts for synthesis of higher alcohols from syngas

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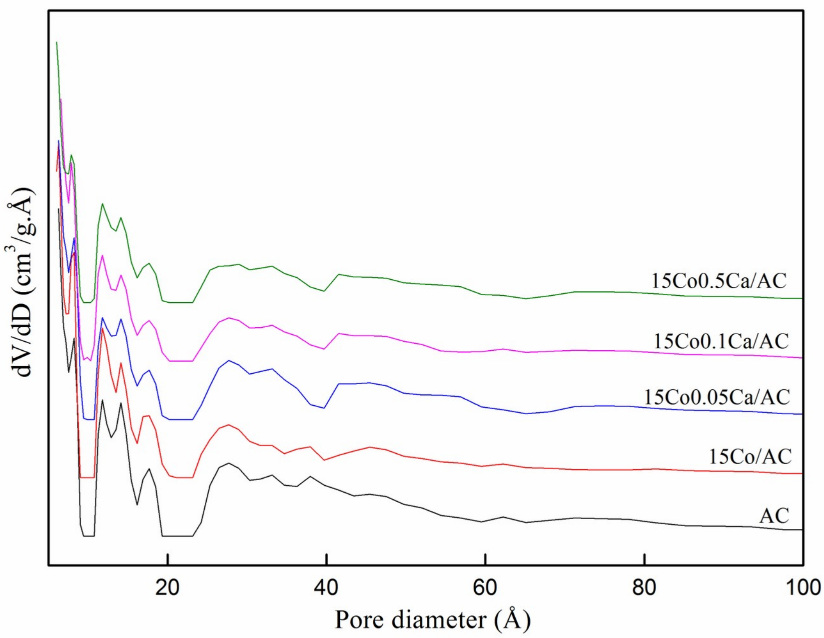

Figure S1. Pore size distribution curves of the AC support and corresponding catalysts calculated from non-local density functional theory (NLDFT).

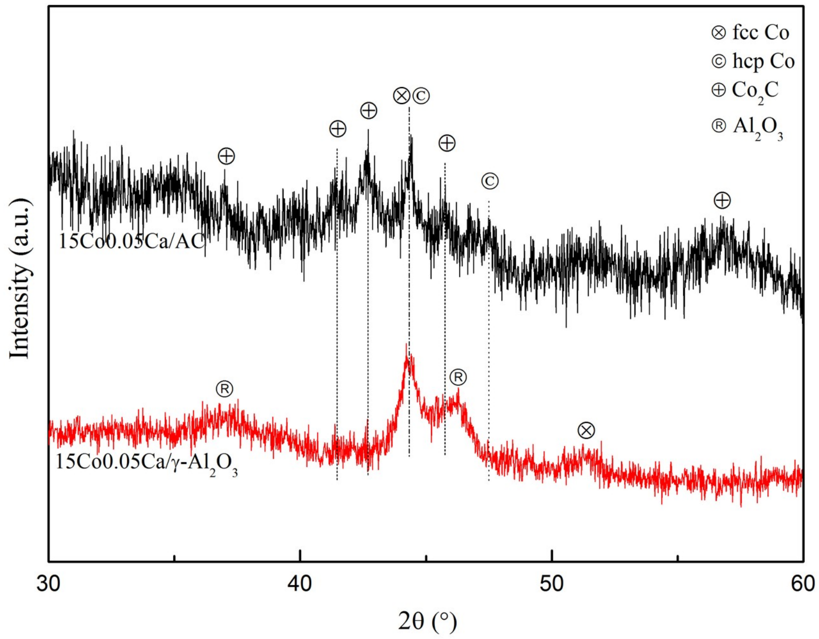

Figure S2. XRD patterns for the spent 15Co0.5Ca/AC and 15Co0.5Ca/γ-Al_2O_3 catalysts.

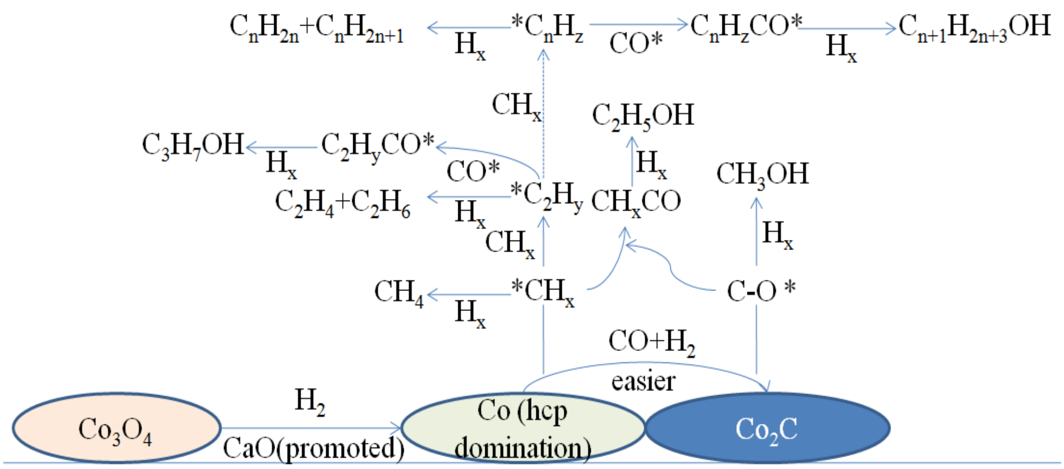

Figure S3. Scheme of CO hydrogenation reaction pathway over the 15Co-xCa/AC catalysts.

Table S1 The performance of CO hydrogenation over the 15Co0.5Ca/AC and 15Co0.5Ca/γ-Al_2O_3 catalysts [a]

Catalyst	CO Conversion /%	Selectivity/C %				Alcohol Yield, %	Alcohol distribution/wt %	
		HC_{1-4} [b]	CO_2	HC_{5+} [c]	ROH [d]		C_1OH [e]	$C_{2+}OH$ [f]
15Co0.5Ca/AC	24.7	48.0	1.3	21.4	29.3	7.2	4.2	95.8
15Co0.5Ca/γ-Al_2O_3	16.5	56.9	0.6	34.1	8.4	1.4	6.2	93.8

[a] Reaction conditions: P = 3.0 MPa, T = 493 K, GHSV = 900 h^{-1}.

[b] Hydrocarbons with carbon numbers of 1-4.

[c] Hydrocarbons with carbon numbers more than 4.

[d] Alcohols with carbon numbers of 1-18.

[e] CH_3OH

[f] Alcohols with carbon numbers more than one..